

HUMAN-VS-AI EXPLAINER

How Human Writing Differs From AI Writing — And How To Fix AI-Sounding Text

This guide has three jobs. First, explain *why* AI writing sounds the way it does. Second, show the *specific, observable* ways human writing differs. Third — and most important — give you *repeatable techniques* you can apply to any draft tonight.

Part 1: Why AI Writing Sounds Robotic

Large language models are next-word prediction machines

A large language model (LLM) — the technology behind ChatGPT, Claude, Gemini, and similar tools — is, at its core, a very sophisticated autocomplete. "Large language model" just means a computer program trained on an enormous amount of text so it can guess what words tend to follow other words.

Here is the whole trick in one sentence: *the model reads everything you have written so far and produces a ranked list of probabilities for what the next word (technically, the next "token" — a word or word-piece) should be, then picks one.* Then it repeats, word after word, until it stops.

If the input is "For lunch today, I ate a bowl of __," the model rates "soup," "cereal," or "rice" as high-probability and "spiders" or "gravel" as very low-probability. Left to its default settings, it leans toward the safe, high-probability choice almost every time.

There is a setting called *temperature* that controls how adventurous the model is. Low temperature = it almost always grabs the single most likely word (predictable, repetitive). High temperature = it takes more chances (more varied, but sometimes nonsense). At a temperature of zero, the model becomes fully deterministic — feed it the same prompt and it returns the same words every time. Most consumer chatbots run at a moderate temperature tuned for coherence and safety, not surprise. That tuning is one reason the output feels smooth but bland.

Why this produces "average" prose

Because the model is trained to predict the **most likely** continuation, it naturally gravitates to the middle of everything: the most common phrasing, the most conventional structure, the least risky word. It is, in effect, writing the statistical average of everything it has read.

Think of it like a singer who only ever hits the exact center of every note. Technically flawless. Emotionally dead. Human writers bend notes — they choose the odd word, the abrupt sentence, the personal aside — precisely because those are **not** the most probable choices. That "off-center" quality is a lot of what we mean by voice.

The role of RLHF and the "assistant" tone

There is a second, less obvious reason AI writing has a recognizable flavor. After the raw model is trained, companies fine-tune it using a process called ***RLHF — Reinforcement Learning from Human Feedback.*** In plain terms: human reviewers are shown different answers the model could give and they rank which ones they prefer. Those rankings train a "reward model" that then steers the LLM toward the kind of output humans rewarded.

RLHF is the reason ChatGPT behaves like a polite, eager, slightly over-explaining assistant instead of a raw autocomplete engine. It is also, researchers now believe, part of why models overuse certain words. A 2025 study presented at the 31st International Conference on Computational Linguistics (Tom Juzek & Zina Ward, "Why Does ChatGPT 'Delve' So Much?") identified 21 "focal words" — including **delve**, **intricate**, and **underscore** — whose spike in scientific abstracts tracks LLM usage, and found that model testing "is consistent with RLHF playing a role" in that overuse. In an online experiment emulating the RLHF ranking process, participants reacted especially negatively to **delve** compared with other buzzwords — a hint that the "why" is still not fully settled.

There is also a fascinating, widely discussed theory about the word "delve" specifically: much RLHF annotation work is outsourced to English-speaking workers in regions like Africa, and in Nigerian formal/business English, "delve" is a normal, common word. The theory (raised by journalist Alex Hern and amplified by technologist Simon Willison) is that annotator language preferences got baked into the model's rewards. It is a plausible, unproven theory — treat it as an illustrative story, not established fact. The deeper point holds regardless: ***the model's quirks come from human choices during training, which is why they cluster into a recognizable "house style."***

Part 2: Perplexity — How Predictable the Words Are

What it means

Perplexity is a measure of how *surprised* a language model is by a piece of text. The word shares a root with "perplexed" — confused, puzzled. Low perplexity = the words are exactly what the model expected. High perplexity = the words are surprising.

A simple example (borrowed from Pangram Labs' explainer):

- **Low perplexity:** "For lunch today, I ate a bowl of *soup*."
- **High perplexity:** "For lunch today, I ate a bowl of *spiders*."

"Spiders" is a shocking, low-probability ending, so it scores high perplexity.

Why AI text scores low and human text scores higher

AI text tends to be low-perplexity *by construction* — the model literally wrote the words it found most probable, so of course they look expected. Human writing has more spikes of surprise, because humans pick words for reasons a probability model does not optimize for: a specific memory, a joke, a rhythm, a regional expression, a piece of context only they have.

What raises perplexity (what humans do)

- **Specificity:** naming the exact thing instead of the general category ("a dented '98 Corolla" vs "a vehicle").
- **Unexpected but apt word choice:** the slightly-wrong-on-purpose word that turns out to be exactly right.
- **Personal reference:** something from your actual life that a model could not have predicted.

Honest caveat: perplexity is real and measurable, but it is not a clean line between human and machine. Formal, careful human writing (legal text, academic abstracts, and often the writing of non-native English speakers) is *also* low-perplexity — which is the root of the detector-bias problem covered in Part 7.

Part 3: Burstiness — Variation in Rhythm

What it means

Burstiness describes how much a text *varies* — in sentence length, in structure, and in that perplexity measure — across a whole document. High burstiness: short punchy sentences slammed against long winding ones. Low burstiness: every sentence roughly the same length and shape, a steady metronomic beat.

Here is low burstiness you can hear:

> "I have a dog. My dog's name is Tim. Tim looks a bit scruffy. Tim's ears stick out."

Four sentences, nearly identical length and shape. It drones.

Why humans vary and AI stays in a narrow band

Human writers instinctively break up rhythm. We get bored of our own cadence; short-term memory nudges us not to repeat the same shape twice in a row. Models, by contrast, tend to settle into a consistent, even rhythm — the same tidy sentence length, paragraph after paragraph.

The single most-quoted illustration of what good burstiness sounds like comes from copywriter Gary Provost:

> "This sentence has five words. Here are five more words. Five-word sentences are fine. But several together become monotonous. Listen to what is happening. The writing is getting boring. The sound of it drones. It's like a stuck record. The ear demands some variety. Now listen. I vary the sentence length, and I create music. Music. The writing sings..."

That passage *is* the technique. Read it aloud and you feel the difference variation makes.

The honest truth about burstiness (read this carefully)

Burstiness is often sold as "the single clearest signal separating human from AI." That claim deserves real skepticism.

The term did not originate in AI detection at all. It comes from 1990s information-retrieval and corpus-linguistics research — Kenneth Church and William Gale's work on how word occurrences deviate from a random (Poisson) distribution. Their memorable phrasing: the Poisson model assumes "lightning is unlikely to strike twice in a single document," but in real text, "when it rains, it pours" — words cluster in bursts. It was a statistical description of word-occurrence patterns, not a lie-detector for machines.

Burstiness became an *AI-detection* concept in January 2023, when GPTZero's first public model framed detection around perplexity plus burstiness. But here is the part the marketing rarely mentions: **GPTZero itself has publicly moved on.** In its own words (verbatim, from its support center): "As of autumn 2023, GPTZero no longer uses perplexity and burstiness for its AI detection because we migrated to a deep-learning based architecture."

And peer-reviewed work is blunt about the limits. A 2023 academic study (the "Counter Turing Test," Chakraborty et al., presented at EMNLP 2023) investigated perplexity and burstiness as detection signals and concluded that "the text produced by newer LLMs is nearly indistinguishable from human-written text from a statistical perspective."

So how should you treat burstiness? As excellent *craft advice* and a *weak forensic signal*. Varying your sentence rhythm genuinely makes writing better and more human to read — that part is true and worth doing. But the idea that a burstiness score reliably unmask a machine is largely detector-company framing that the evidence does not fully support. Do it because it improves the writing, not because it beats a scanner.

Part 4: The Specific Tells of AI Writing

These fall into three buckets: **vocabulary, structure, and texture.** No single tell proves anything — careful humans use all of these sometimes. It is *convergence* (several tells stacking up in one short passage) that signals machine-default writing.

Tell #1: Overused vocabulary (the strongest evidence base)

This is the most rigorously studied tell. A landmark study — Kobak et al., "Delving into LLM-assisted writing in biomedical publications through excess vocabulary," published in *Science Advances* in July 2025 — analyzed more than 15 million PubMed abstracts from 2010–2024. The researchers borrowed a method from pandemic "excess mortality" studies: they measured how much certain words *exceeded* their expected frequency after ChatGPT arrived.

The findings are striking. Certain "style words" (as opposed to content words) exploded in frequency in 2024. The standout: *"delves"* appeared at roughly *28 times* its pre-ChatGPT baseline (a frequency ratio of $r = 28.0$). "Underscores" hit about $14\times$ ($r = 13.8$), "showcasing" about $11\times$ ($r = 10.7$). In the authors' own words, "this excess word analysis suggests that at least 13.5% of 2024 abstracts were processed with LLMs. This lower bound differed across disciplines, countries, and journals, reaching 40% for some subcorpora." Crucially, the 2024 excess vocabulary was "almost entirely" style words (like *intricate*, notably*), whereas the COVID-era vocabulary spike was content words (like *respiratory*). That is the fingerprint of a change in *style*, not subject.

A usable flagged-word list. Drawn from Kobak et al. (2025), Juzek & Ward (2025), Liang et al. (2024), and related corpus studies. These are not "banned" — they are words to *notice*, because a cluster of them signals default-AI phrasing:

**Verbs:* delve, delve into, underscore, showcase, highlight, emphasize, foster, leverage, navigate, unlock, harness, embark, elevate, boast, garner, align, streamline, tailor, resonate, surpass, encompass, underpin, bolster.

**Adjectives:* intricate, pivotal, crucial, comprehensive, meticulous, robust, seamless, notable, noteworthy, invaluable, commendable, groundbreaking, nuanced, multifaceted, holistic, vibrant, rich, vast, ever-evolving, ever-changing.

**Nouns:* tapestry, realm, landscape, journey, testament, insights, nuance, interplay, cornerstone, beacon, treasure trove, game changer, wealth (of information), paradigm.

**Adverbs:* notably, importantly, significantly, meticulously, seamlessly, ultimately, arguably.

**Phrases and connective tissue:* "in today's world," "in the ever-evolving landscape of," "it's important to note," "it's worth noting," "when it comes to," "at the end of the day," "plays a crucial/pivotal role," "a testament to," "navigating the complexities of," "unlock the potential," "dive deeper," "rich tapestry," "stand as a testament," "in conclusion," "in the realm of."

****Important honesty flag:**** this research is **domain-specific** (mostly scientific abstracts) and the authors themselves warn it is "not a silver bullet." A human can legitimately write "delve." Non-native English speakers may use these words normally (again, "delve" is standard Nigerian business English). The list is a **smell test**, not a verdict.

Tell #2: Structural habits

- ****Uniform sentence length.**** The metronomic beat from Part 3.
- ****The rule of three, everywhere.**** "Efficient, scalable, and reliable." "Plan, execute, and optimize." Humans use three-item lists occasionally; AI reaches for the balanced triplet as a default. The tell is **frequency**, not the device itself.
- ****"Not only... but also" and "It's not just X, it's Y."**** This "negation pivot" or contrastive construction is catnip to LLMs — a formula for fake profundity. "It's not just a phone, it's a lifestyle."
- ****Em-dash overuse.**** ChatGPT loves the em dash (—), often several per paragraph, slammed between clauses that a comma or period would serve better. ****Strong caveat:**** the em dash is a legitimate, beloved punctuation mark used by great writers for centuries (Emily Dickinson, anyone). It is not itself proof of AI. The tell is **density and mechanical use**, not the mark. (OpenAI's Sam Altman acknowledged the quirk publicly in late 2025, and you can now instruct models to avoid it.)
- ****Paragraph uniformity.**** Every paragraph follows the same recipe: claim, example, restatement. Same shape, over and over.
- ****The "in conclusion" wrap-up.**** A closing paragraph that dutifully summarizes what was just said, like a five-paragraph school essay.

Tell #3: Texture — the "too smooth, says nothing" problem

- ****Over-hedging.**** "It is important to consider," "there are many factors," "results may vary." The writing sounds cautious and authoritative while committing to nothing.
- ****Generic abstraction over concrete detail.**** AI writes "a feast for the senses"; a human writes "the fish stall smelled of ice and copper." AI names the **category**; humans name the **thing**.
- ****No point of view.**** Ask an AI for an opinion and it often gives you both sides, perfectly balanced, ending on "ultimately, it depends." Humans take sides.
- ****Excessive smoothness / no rough edges.**** Paradoxically, writing that is **too** clean — no fragments, no asides, no small imperfections — reads as machine-made. Real human writing has texture: a dropped-in aside, a one-word sentence, a rhetorical question, an admission of doubt.

Part 5: What Human Writing Actually Does Differently

1. Sentence-length variation

Humans mix a two-word sentence with a forty-word sentence on purpose. It creates rhythm and controls the reader's breathing. (See the Gary Provost passage in Part 3 — that is the whole lesson in one paragraph.)

2. Concrete specifics over generalities

This is the highest-leverage difference. Compare:

- **Generic (AI-ish):** "The café had a cozy atmosphere and friendly staff."
- **Concrete (human):** "The café had six tables, a barista who remembered my order after one visit, and a radiator that clanked every twenty minutes."

The specifics do double duty: they raise perplexity (a model can't predict "clanked every twenty minutes") and they *prove you were actually there.*

3. Personal voice, opinion, and asides

Humans interrupt themselves. They editorialize. They admit uncertainty ("I'm still not sure this is right, but..."). They drop in a parenthetical joke. These moves are the opposite of writing the statistical average — they are irreducibly personal.

4. Rhythm, fragments, and intentional rule-breaking

Starting a sentence with "And." A deliberate fragment. For emphasis. A sentence that runs long on purpose because the idea keeps unspooling and you want the reader to feel a little breathless before it finally lands. Humans break the rules they know; AI defaults to the rules.

Part 6: Techniques to Fix AI-Sounding Text

Here is an ordered edit pass. Do them in this sequence — structure first, words last.

The ordered checklist

1. **Read it aloud first.** This is the single most reliable test. Your ear catches the drone, the hedging, and the fake-profound pivots faster than your eye. Mark every spot where you get bored.
2. **Vary the rhythm.** Go paragraph by paragraph. Find runs of same-length sentences. Split one. Merge two. Add a short one. *Three words. Just like that.*
3. **Cut the stock phrases and tell-words.** Search-and-destroy "it's important to note," "in today's world," "a testament to," "delve," "tapestry," "navigate the complexities." Replace with plain, direct language — or nothing.
4. **Inject specificity.** For every vague statement, ask: *what's the actual example?* Add one detail per section that only you could know — a name, a number, a date, a smell, a moment.
5. **Add a point of view.** State what you actually think. Cut the both-sides hedge. If the AI wrote "there are pros and cons," pick one and defend it.
6. **De-smooth it.** Add one aside, one fragment, or one honest admission of doubt. Break one rule on purpose.
7. **Kill the wrap-up.** Delete the "In conclusion, ..." paragraph that just repeats everything. End on something sharper.

Before/After examples

Example A — a product description

> **BEFORE (AI-sounding):** In today's fast-paced world, our innovative water bottle is a game changer that seamlessly combines cutting-edge design with unparalleled functionality, making it the perfect companion for your daily journey.

> **AFTER (human):** It keeps water cold for 24 hours. I've left it in a hot car all day and the ice was still there at dinner. The lid doesn't leak in a gym bag — I checked, twice, after the first one I owned soaked a laptop.

What changed: Cut "in today's fast-paced world," "game changer," "seamlessly," "cutting-edge," "unparalleled," "journey." Replaced abstract praise with concrete, testable claims and a personal mishap. Sentence lengths now vary.

****Example B — a blog intro****

> ****BEFORE:**** When it comes to productivity, it's important to note that there are many factors to consider. Building good habits can play a pivotal role in unlocking your full potential and navigating the complexities of modern work life.

> ****AFTER:**** Most productivity advice is useless to me by 3 p.m. That's when my focus collapses and I start reorganizing my desk instead of working. So I stopped chasing "habits" and changed one thing: I put my phone in another room. That's it. My afternoons came back.

What changed: Killed three hedge phrases and four tell-words. Added a specific time, a specific failure, and one concrete action. The voice now has an opinion and an admission of weakness.

****Example C — a professional email (note for non-native writers)****

> ****BEFORE:**** I am writing to delve into the matter we discussed and to underscore the crucial importance of aligning our comprehensive strategy moving forward.

> ****AFTER:**** Following up on our chat. The key thing is that our two teams pick the same launch date — right now marketing says June and sales says August, and that gap is the whole problem.

What changed: Removed "delve," "underscore," "crucial," "comprehensive," "aligning," "moving forward." Named the actual conflict (June vs August). Shorter, clearer, and more direct — which is **better** business writing, not just less AI-like. (If English is your second language, notice this fix makes the writing **simpler**, not fancier. The goal is your real voice, not more impressive vocabulary.)

****Example D — narrative / personal essay****

> ****BEFORE:**** My grandmother's kitchen was a rich tapestry of sights and smells that served as a testament to her enduring love for the culinary arts.

> **AFTER:** My grandmother's kitchen always smelled like burnt sugar and bleach. She'd start caramel, forget it, then scrub the pan while the smoke alarm went off. She was not a good cook. I miss it violently.

What changed: Cut "rich tapestry," "testament," "enduring love," "culinary arts" — the whole bouquet of AI abstractions. Replaced with two specific smells, a repeated behavior, a blunt opinion ("not a good cook"), and an unexpected emotional turn.

A note on prompting (fix it upstream)

You can reduce the edit burden by prompting better. Two moves that genuinely help:

- **Feed the model your voice.** Paste 2–3 paragraphs of your own real writing and say: "Match this voice, including the short sentences and the asides in parentheses."
- **Ask for uneven rhythm explicitly.** "Vary sentence length dramatically; include at least one fragment and at least one sentence over 30 words. Avoid the words delve, tapestry, underscore, and the phrase 'it's important to note.'"

Prompting won't get you all the way. The details, the opinions, the real examples — those still have to come from you.

Part 7: How AI Detectors Work (And Why You Shouldn't Trust Them)

What they actually measure

Early AI detectors (GPTZero's first model, and academic tools like DetectGPT) were built on exactly the two concepts from Parts 2 and 3: **perplexity** (is the text low-surprise?) and **burstiness** (does the rhythm stay in a narrow band?). The logic: AI text is uniformly low-perplexity and low-burstiness; human text has spikes and variation. Newer commercial detectors mostly use trained deep-learning classifiers instead, but the underlying intuition is the same.

Why they fail, and who gets hurt

****Problem 1 — They flag famous human texts.**** Because models are trained to make familiar documents low-perplexity, perplexity-based detectors have flagged the U.S. Declaration of Independence and passages of the Bible as "AI-generated." The text is so widely reproduced online that the model finds it completely unsurprising — indistinguishable, to the detector, from machine output.

****Problem 2 — They are biased against non-native English writers.** This is the most important limitation for this document's readers.** A 2023 Stanford study (Liang et al., published in the journal *Patterns*) tested seven detectors on essays written by non-native English speakers and found they "misclassified over half of the TOEFL essays as 'AI-generated' (average false positive rate: 61.22%)," while essays by native US students were classified accurately. All seven detectors **unanimously** flagged 18 of the 91 essays (19.78%) as AI, and 97.8% were flagged by at least one detector. The reason is structural: non-native speakers often write with simpler, more common vocabulary and standard sentence structures — exactly the low-perplexity pattern detectors associate with machines. In the authors' words, the detectors' design "inherently discriminates against non-native authors." Tellingly, when the researchers prompted ChatGPT to enrich the same essays' vocabulary, the false-positive rate fell from 61.3% to 11.6% — proving the tools were really just penalizing simpler English. This has real consequences: students falsely accused of cheating, with visa and disciplinary stakes.

****Problem 3 — Independent testing shows they're just not reliable.**** A peer-reviewed 2023 study (Weber-Wulff et al., *International Journal for Educational Integrity*) tested 14 detection tools — 12 publicly available tools plus two commercial systems (Turnitin and PlagiarismCheck) — and concluded they are "neither accurate nor reliable (all scored below 80% of accuracy and only 5 over 70%)." The tools were easily defeated by manual editing, paraphrasing, or machine translation, and they leaned toward calling AI text "human" (false negatives). The large-scale RAID benchmark (Dugan et al., ACL 2024) similarly found that "current detectors are easily fooled by adversarial attacks, variations in sampling strategies, repetition penalties, and unseen generative models."

****Problem 4 — The newest models are increasingly indistinguishable.**** Vendors claim 99%+ accuracy, but those numbers come from testing against **raw, unedited** output from older models. GPT-5, Claude 4.5, and Gemini 2.5 produce more varied, higher-perplexity text than their predecessors, and multiple 2025–2026 evaluations report detector accuracy slipping and detectors reading current-model text as human. Detector accuracy also collapses on short texts (GPTZero itself notes false-positive rates rise sharply under ~300 words). Note that these last figures come largely from detector companies and AI-"humanizer" companies, both of which have commercial incentives — treat exact percentages as directional, not gospel.

The honest bottom line on detectors

****"Guaranteed undetectable" is a claim you cannot trust, and chasing it is the wrong goal.**** The statistical gap between good human writing and good AI writing is narrowing, detectors are unreliable in both directions (false accusations *and* missed AI), and the whole thing is an arms race where the evasion tools currently have the upper hand.

So don't write to beat a scanner. Write to sound like a specific human — you. That goal is stable, it makes your writing genuinely better, and it doesn't evaporate the next time a model or a detector updates.

An ethical note

This document is about *voice and craft* — making writing clear, specific, and genuinely yours. It is not a manual for deceiving a teacher, a journal, an employer, or a platform about who or what wrote something. If you're a student or a researcher, follow your institution's disclosure rules. Using AI as a drafting assistant and then rewriting in your own voice is different from passing off unedited machine text as your own original work, and different again from evading a required disclosure. Keep the line clear.

One-Page Cheat Sheet

****WHY AI SOUNDS LIKE AI****

- It predicts the most likely next word → smooth, average, safe prose.
- Fine-tuning (RLHF) gives it a recognizable polite "assistant" house style and favorite words.

****THE TWO MEASURABLE DIFFERENCES****

- **Perplexity** = how surprising the words are. AI = low (predictable). Humans = higher (specific, unexpected).
- **Burstiness** = how much sentence rhythm varies. AI = flat. Humans = spiky. (Great craft advice; weak as a lie-detector — even GPTZero dropped it in 2023.)

****TOP AI TELL-WORDS TO CUT****

delve, tapestry, realm, underscore, showcase, leverage, navigate, pivotal, crucial, intricate, meticulous, seamless, robust, testament, landscape, journey, "it's important to note," "in today's world," "not just X, it's Y," "a testament to," "navigate the complexities."

(These are smell tests, not proof. Careful and non-native writers use them legitimately.)

****STRUCTURAL TELLS****

Uniform sentence length · rule-of-three everywhere · em-dash pileups · "not only... but also" · every paragraph the same shape · a summary "in conclusion" ending.

****THE 7-STEP FIX (structure first, words last)****

1. Read it aloud — mark the boring parts.
2. Vary sentence length. Split, merge, add a short one.
3. Delete stock phrases and tell-words.
4. Add one specific detail per section only you could know.
5. State a real opinion; cut the both-sides hedge.
6. De-smooth: add an aside, a fragment, or an honest doubt.
7. Kill the "in conclusion" wrap-up. End sharp.

****THE ONE-LINE TEST****

After reading a paragraph, can you restate one concrete fact? If not, it's AI-smooth and empty. Add specifics.

****ON DETECTORS****

Unreliable. Biased against non-native English writers (61% average false-positive rate in the Stanford study). Beaten by light editing. Newer models are hard to detect at all. ****Don't write to pass a detector — write to sound like you.****

****THE GOLDEN RULE****

Add the one detail only you would know. That single move does more than any word-swap.